MPSA 2026 - estimation results for Chad

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2026-04-16

Dichotomized treatment version: this PDF shows, in order:

1. defining treatment
2. balance of covariates
3. Kernel weighting (potentially our main result)
4. probably the appendix: all other types of matching and weighting, with balance tables and PDFs of covariates

- naive
- with controls
- matching on covariates
- matching on propensity scores
- weighting on propensity scores

5. table and graph of all results
6. sensitivity analysis

Each of the matching/weighting includes two options: Match() vs Matchit(); manual vs weightit(); how to choose which?

1. Set up

2. Data cleaning

Treatment is disappearances and homicides by the state; calculated as the yearly rate by 10,000 inhabitants of cases for the period between 2003 and 2011 per municipality

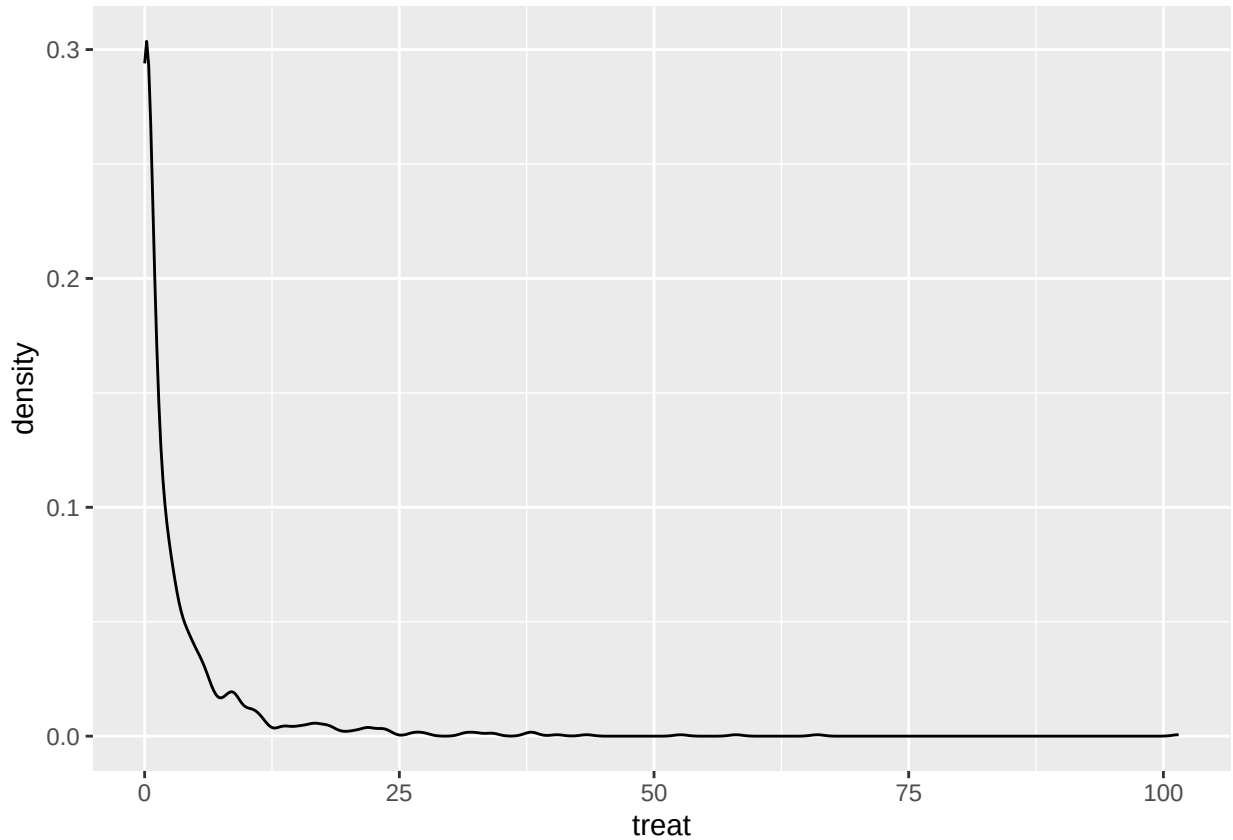
We have several options for the exact calculation of treatment:

1. dichotomized, where any amount of violence is 1
2. cutoff, where a certain rate is considered 1
3. continuous - but we can't do balance tables/matching/weighting
4. extremes, of no violence versus high violence

2.1 choosing treatment and covariates

2.2 Covariates

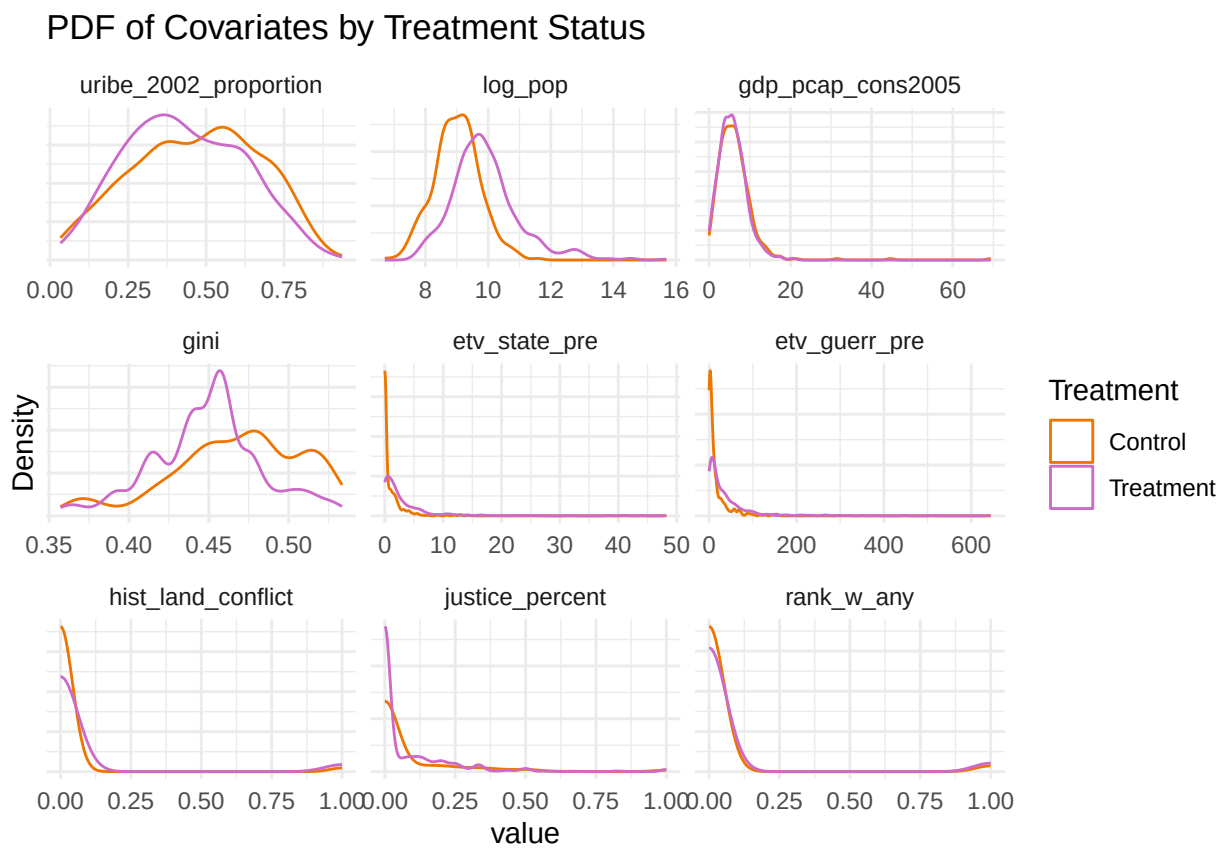
A density plot of the yearly rate of cases per 10,000 per municipality; followed by the exact number and proportion of control municipalities




```
##   0
## 366
```

```
## [1] 0.3638171
```

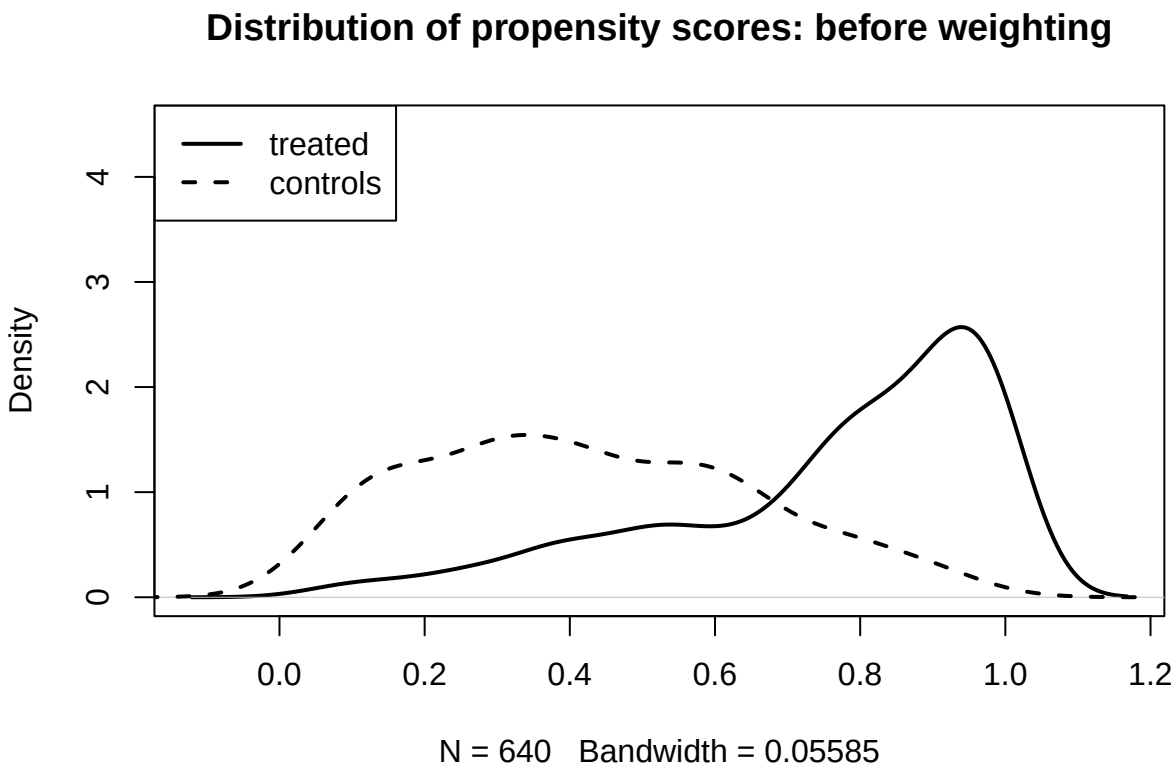
2.2.1 Covariate balance

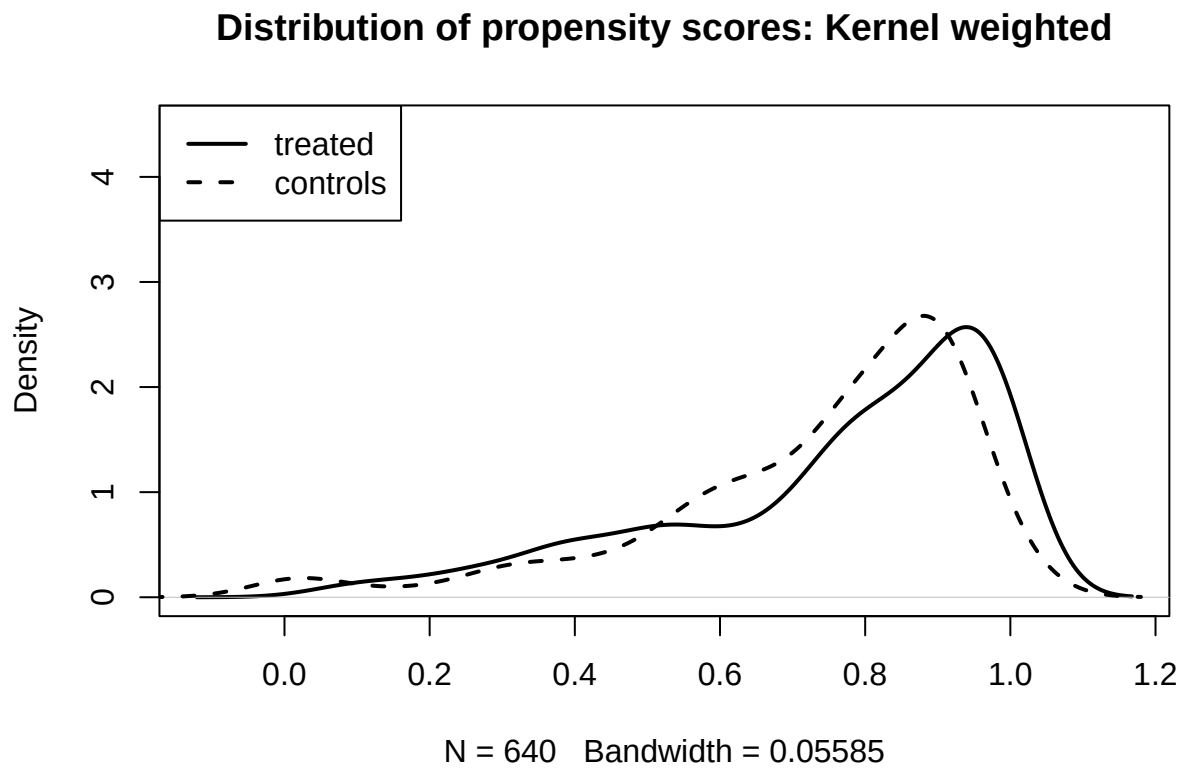


	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.430	0.468	0.004	0.005
log_pop	9.899	8.998	0.000	0.000
gdp_pcap_cons2005	5.798	6.399	0.050	0.334
gini	0.449	0.468	0.000	0.000
etv_state_pre	3.142	0.937	0.000	0.000
etv_guerr_pre	37.680	15.411	0.000	0.000
hist_land_conflict	0.069	0.025	0.001	NA
justice_percent	0.076	0.069	0.528	0.000
rank_w_any	0.064	0.038	0.062	0.047

3. Kernel balancing weights

Propensity scores before and after kernel weighting


Distribution of propensity scores: before weighting
treated
controls
Density
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
N = 640 Bandwidth = 0.05585

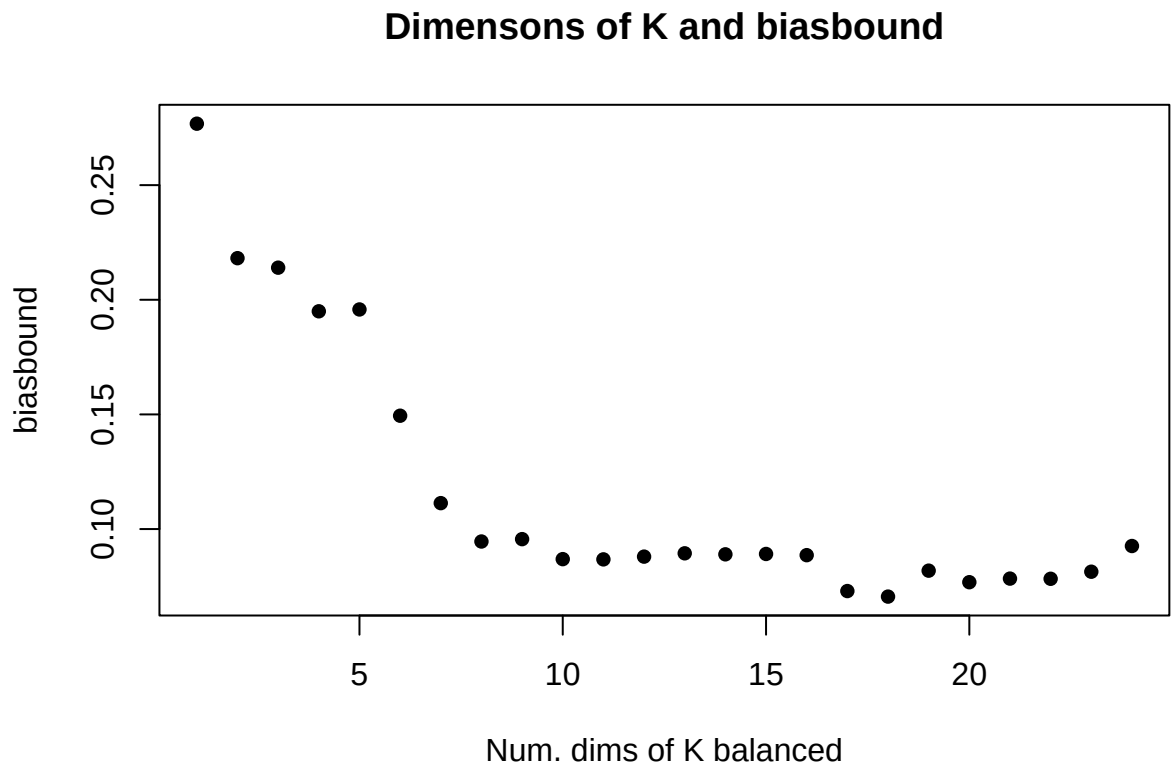


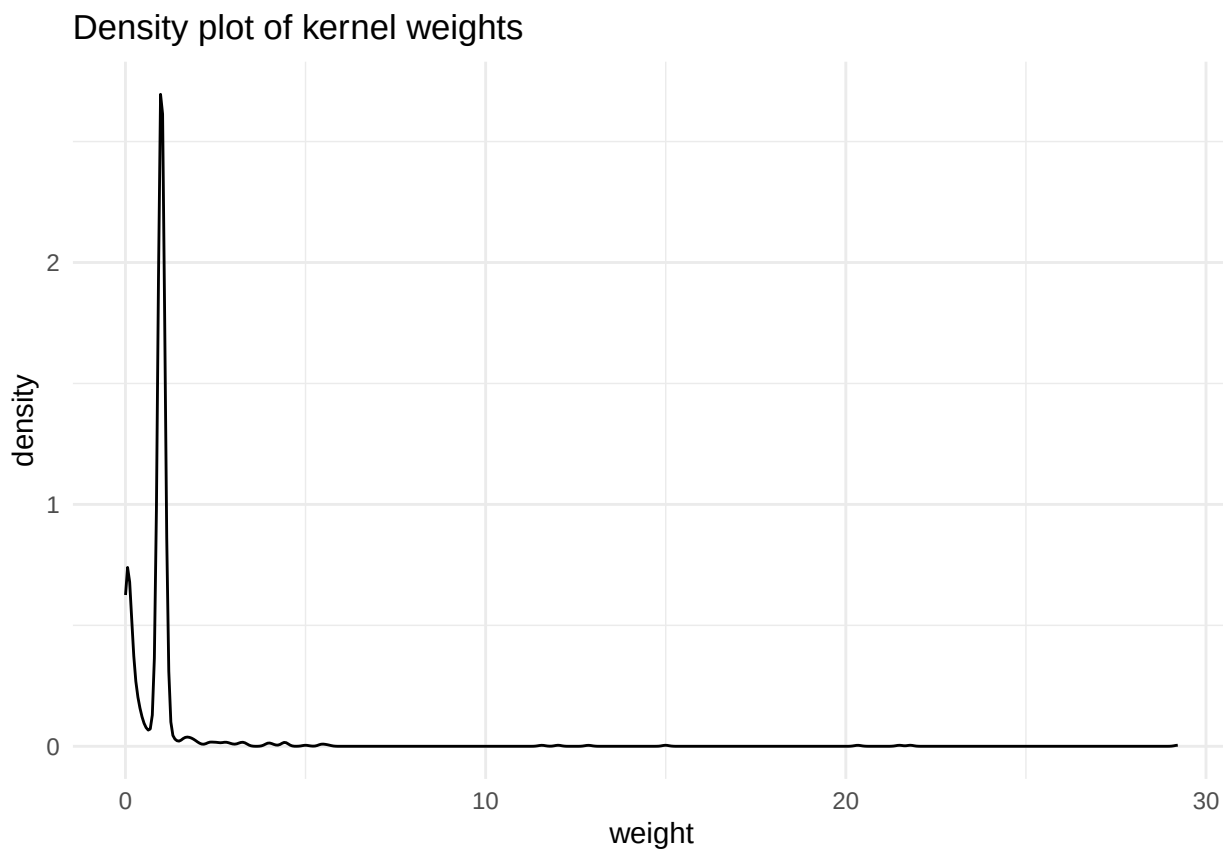
Balance tables, before and after kernel weighting

```
## Balance Measures
##                          Type  M.0.Un  M.1.Un Diff.Un
## uribe_2002_proportion Contin.  0.4677  0.4302 -0.1926
## log_pop               Contin.  8.9984  9.8988  0.9718
## gdp_pcap_cons2005     Contin.  6.3991  5.7980 -0.1367
## gini                  Contin.  0.4677  0.4491 -0.4942
## etv_state_pre         Contin.  0.9374  3.1416  0.5558
## etv_guerr_pre         Contin. 15.4113 37.6798  0.4807
## hist_land_conflict     Binary  0.0246  0.0688  0.0442
## justice_percent       Contin.  0.0688  0.0757  0.0421
## rank_w_any            Contin.  0.0383  0.0643  0.1184
##
## Sample sizes
##     Control Treated
## All     366     640
```

```
## Balance Measures
##                          Type M.0.Adj M.1.Adj Diff.Adj
## uribe_2002_proportion Contin.  0.4384  0.4302  -0.0436
## log_pop               Contin.  9.7159  9.8988   0.1690
## gdp_pcap_cons2005     Contin.  6.7312  5.7980  -0.2860
## gini                  Contin.  0.4501  0.4491  -0.0306
## etv_state_pre         Contin.  2.3720  3.1416   0.1461
## etv_guerr_pre         Contin. 29.4972 37.6798   0.1352
## hist_land_conflict     Binary  0.0637  0.0688   0.0051
```

```
## justice_percent       Contin.  0.0863  0.0757  -0.0699
## rank_w_any            Contin.  0.0591  0.0643   0.0213
##
## Effective sample sizes
##            Control Treated
## Unadjusted  366.       640
## Adjusted     39.54     640
```





Where treatment and control are the most balanced (ebalance convergence) and bias bound is as low as possible

Results:

```
##
## Call:
## lm_robust(formula = trust_plebi_yes ~ treat_type + uribe_2002_proportion +
##     log_pop + gdp_pcap_cons2005 + gini + etv_state_pre + etv_guerr_pre +
##     hist_land_conflict + justice_percent + rank_w_any, data = estimation_nona,
##     weights = weight_model_kbal$w)
##
## Weighted, Standard error type:  HC2
##
## Coefficients:
##                        Estimate Std. Error  t value  Pr(>|t|)   CI Lower
## (Intercept)           1.2540492  0.1031035  12.1630 7.844e-32  1.0517239
## treat_type           -0.0220686  0.0127914  -1.7253 8.479e-02 -0.0471699
## uribe_2002_proportion -0.4498135  0.0335180 -13.4200 7.295e-38 -0.5155876
## log_pop              -0.0025137  0.0048919  -0.5138 6.075e-01 -0.0121134
## gdp_pcap_cons2005    -0.0035713  0.0021068  -1.6951 9.036e-02 -0.0077054
## gini                 -1.1020125  0.1658125  -6.6461 4.949e-11 -1.4273949
## etv_state_pre        -0.0019740  0.0013930  -1.4171 1.568e-01 -0.0047076
## etv_guerr_pre        -0.0001331  0.0001432  -0.9294 3.529e-01 -0.0004141
## hist_land_conflict   -0.0071857  0.0167127  -0.4300 6.673e-01 -0.0399819
## justice_percent       0.0392074  0.0406266   0.9651 3.347e-01 -0.0405162
## rank_w_any            0.0434752  0.0193662   2.2449 2.499e-02  0.0054719
```

```
##                          CI Upper  DF
## (Intercept)             1.4563745 995
## treat_type              0.0030327 995
## uribe_2002_proportion -0.3840393 995
## log_pop                 0.0070860 995
## gdp_pcap_cons2005       0.0005629 995
## gini                   -0.7766301 995
## etv_state_pre           0.0007596 995
## etv_guerr_pre           0.0001479 995
## hist_land_conflict      0.0256105 995
## justice_percent         0.1189310 995
## rank_w_any              0.0814784 995
##
## Multiple R-squared:  0.4853 ,    Adjusted R-squared:  0.4801
## F-statistic: 40.92 on 10 and 995 DF,  p-value: < 2.2e-16
```

4 Other versions

4.1 Naive regression

4.2 Naive + Control regression

4.3 Matching

4.3.1 Matching on covariates - Match() version Throws away non-matched control units

- No replacement and one-to-one matching
- with Match(): closest X_i through normalized Euclidean distance
- BiasAdjust = TRUE: bias correction to adjust for multiple continuous variables
- other things to consider: how many matches, which algorithm (eg Coarse, pre-processing), whether to drop treatment units to remove bias, calipers to remove units with no common support

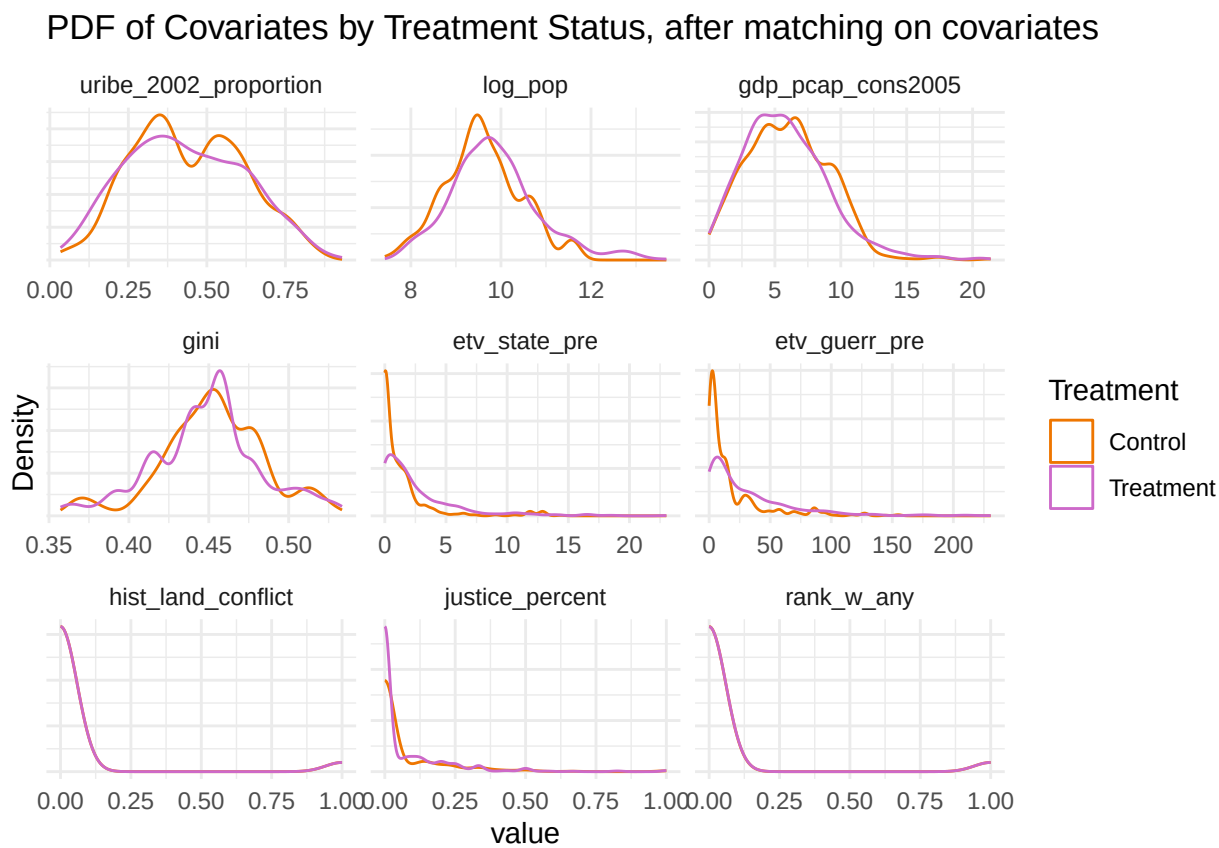
First, creating a matched dataset:

```
##
##   0   1
## 606 606
```

Should we run it multiple times? The results don't change much

Then, checking the balance after the matching

	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.431	0.442	0.006	0.076
log_pop	9.867	9.591	0.000	0.000
gdp_pcap_cons2005	5.798	5.996	0.018	0.012
gini	0.449	0.453	0.000	0.006
etv_state_pre	2.588	1.354	0.000	0.000
etv_guerr_pre	30.552	17.228	0.000	0.000
hist_land_conflict	0.059	0.059	1.000	NA
justice_percent	0.070	0.059	0.000	0.000
rank_w_any	0.060	0.059	0.101	0.843

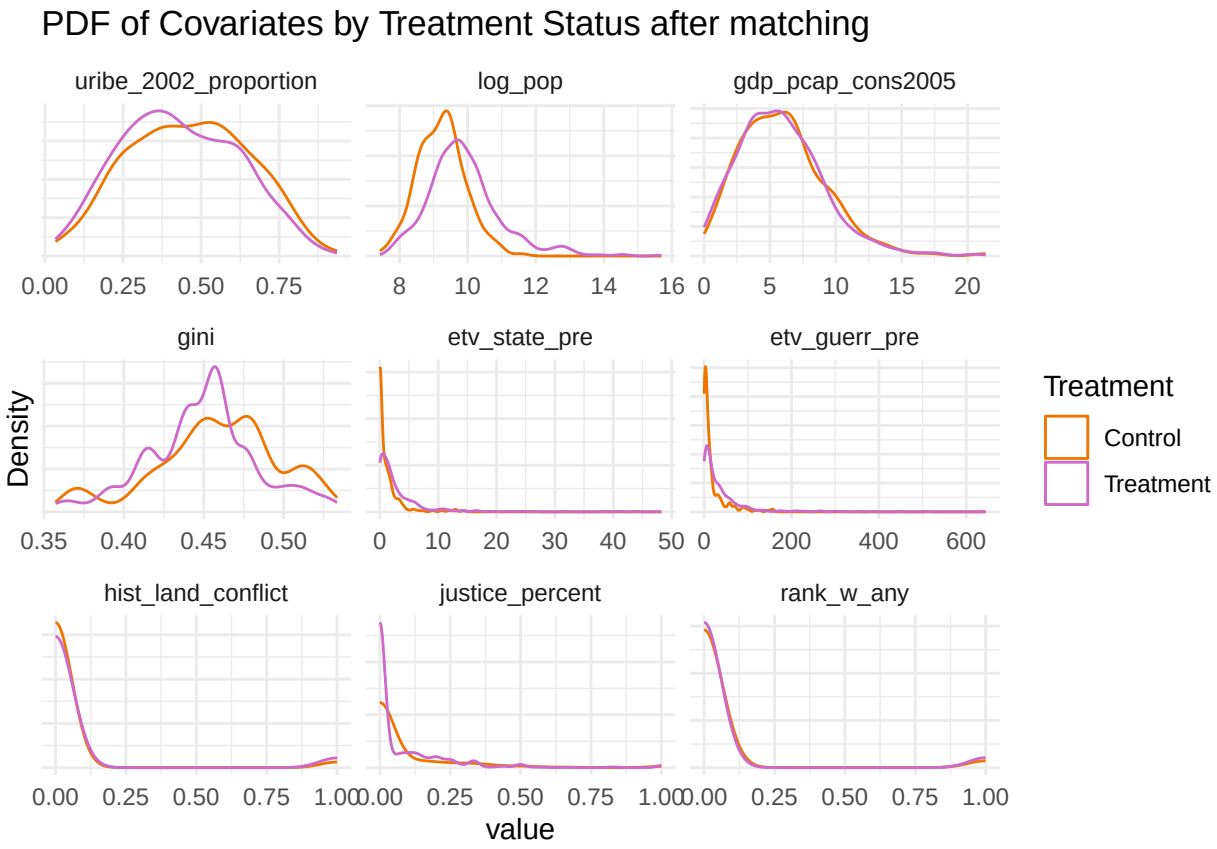


Finally, calculating ATT (how should we?)

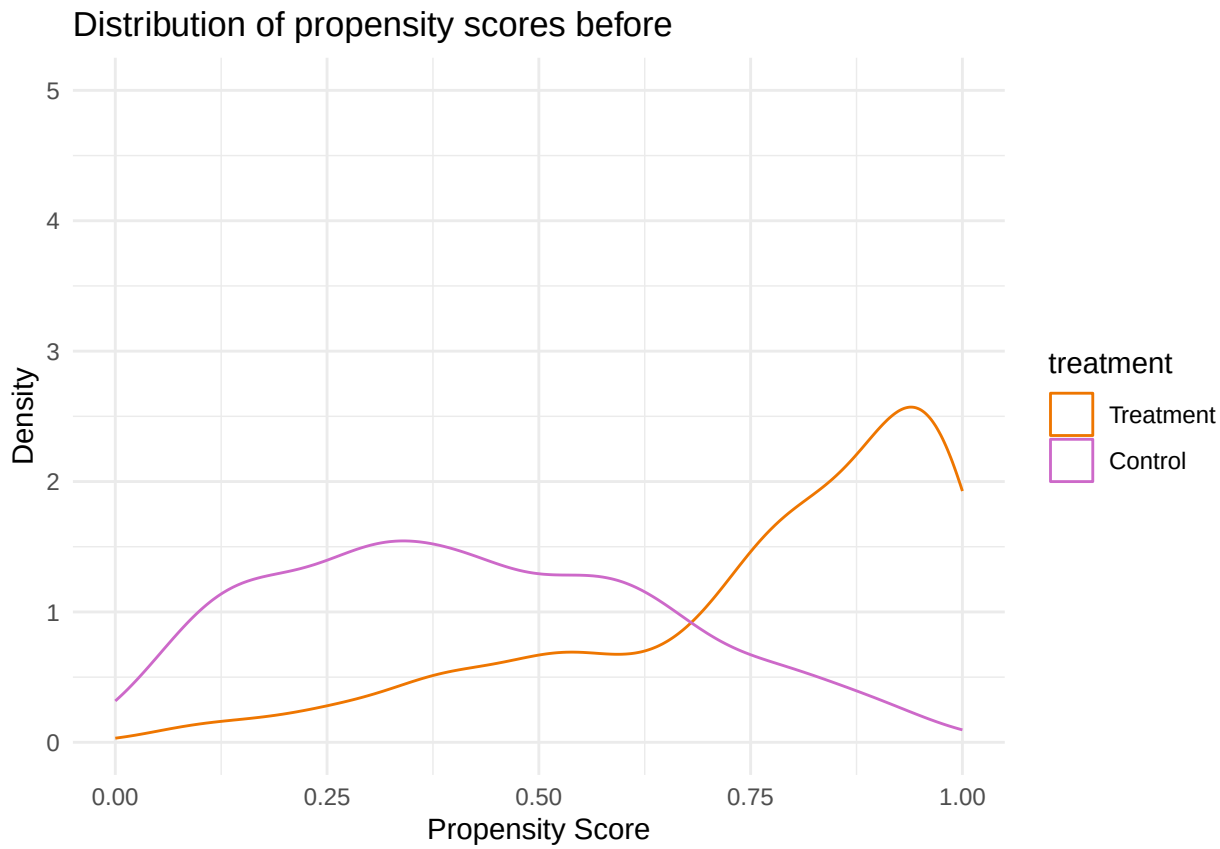
4.3.2 Matching on covariates - MatchIt() version A different way to match on covariates; 1-on-1 nearest neighbor matching on the scaled euclidian distance, with replacement

```
## Balance Measures
##                          Type Diff.Adj KS.Adj
## uribe_2002_proportion Contin.  -0.0631 0.0687
## log_pop               Contin.   0.2893 0.1641
## gdp_pcap_cons2005     Contin.  -0.1175 0.1141
## gini                  Contin.  -0.0940 0.0922
## etv_state_pre         Contin.   0.2766 0.2344
## etv_guerr_pre         Contin.   0.2797 0.2203
## hist_land_conflict     Binary   0.0016 0.0016
## justice_percent       Contin.   0.1189 0.1297
## rank_w_any            Contin.   0.0138 0.0094
##
## Sample sizes
##                      Control Treated
## All                   366.       640
## Matched (ESS)          91.92     640
## Matched (Unweighted)  214.       640
## Unmatched             152.         0
```

```
##
##   0   1
## 214 640
```


PDF of Covariates by Treatment Status after matching
uribe_2002_proportion
log_pop
gdp_pcap_cons2005
gini
etv_state_pre
etv_guerr_pre
hist_land_conflict
justice_percent
rank_w_any
Density
value
Treatment
Control
Treatment

4.3.3 Propensity score matching - Match() version

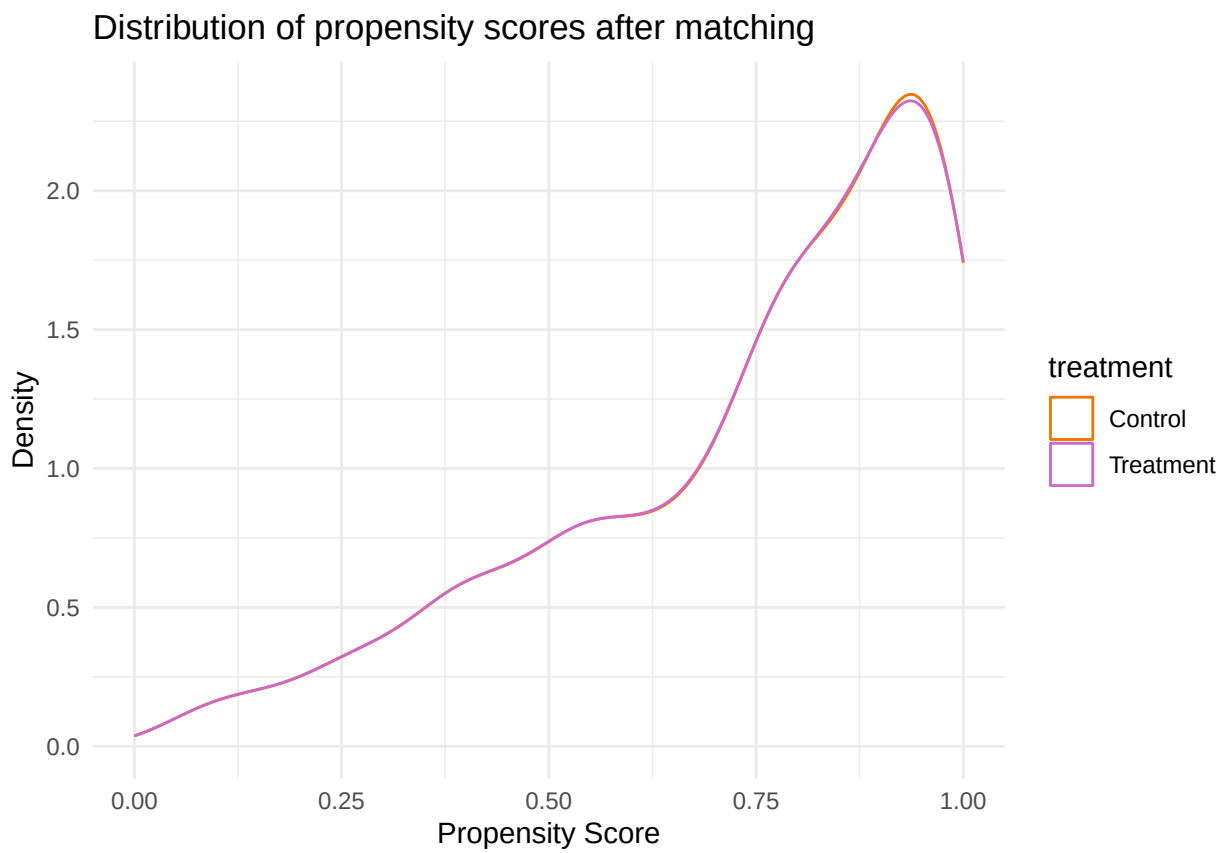


Compare units that have a comparable probability of getting treated to match:

```
##
##   0   1
## 711 711
```

Checking balance after matching:

	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.430	0.512	0.000	0.000
log_pop	9.899	9.532	0.000	0.000
gdp_pcap_cons2005	5.798	6.469	0.000	0.000
gini	0.449	0.454	0.006	0.000
etv_state_pre	3.142	5.248	0.000	0.000
etv_guerr_pre	37.680	32.823	0.044	0.000
hist_land_conflict	0.069	0.034	0.006	NA
justice_percent	0.076	0.051	0.002	0.000
rank_w_any	0.064	0.058	0.643	0.688



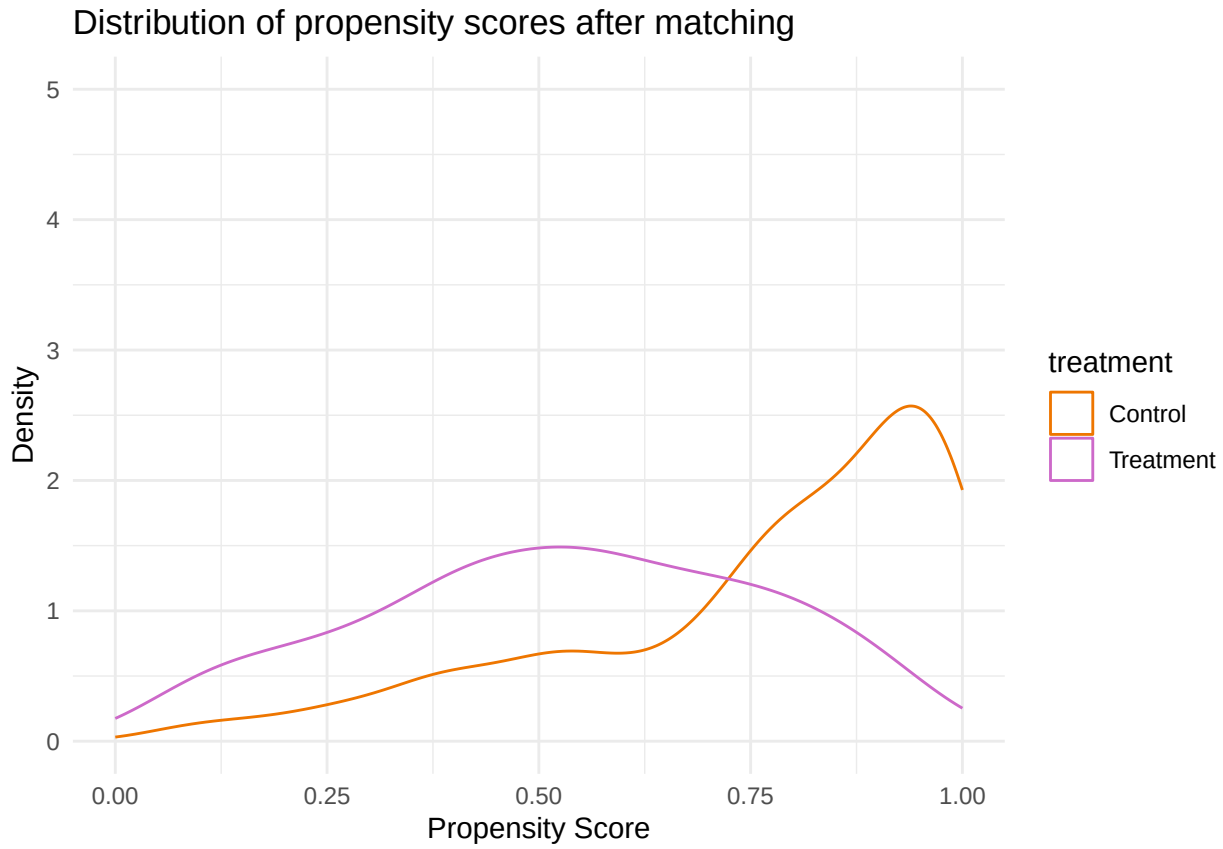
Calculating the ATT:

The Match() propensity scores afterwards are very close, but the data has been significantly manipulated: there are many duplicates of both control and treatment

4.3.4 Propensity score matching - MatchIt() version 1-on-1 nearest neighbor matching on the propensity score, with replacement

```
## Balance Measures
##                           Type Diff.Adj KS.Adj
## distance               Distance   0.0011 0.0812
## uribe_2002_proportion  Contin.  -0.4418 0.3016
## log_pop                Contin.   0.3382 0.2047
## gdp_pcap_cons2005      Contin.  -0.2107 0.1828
## gini                   Contin.  -0.1516 0.1937
## etv_state_pre          Contin.  -0.4024 0.2672
## etv_guerr_pre          Contin.   0.0832 0.1500
## hist_land_conflict      Binary   0.0359 0.0359
## justice_percent        Contin.   0.1559 0.1625
## rank_w_any             Contin.   0.0329 0.0141
##
## Sample sizes
##                      Control Treated
## All                   366.       640
## Matched (ESS)          25.44     640
## Matched (Unweighted)  159.       640
## Unmatched             207.         0
```

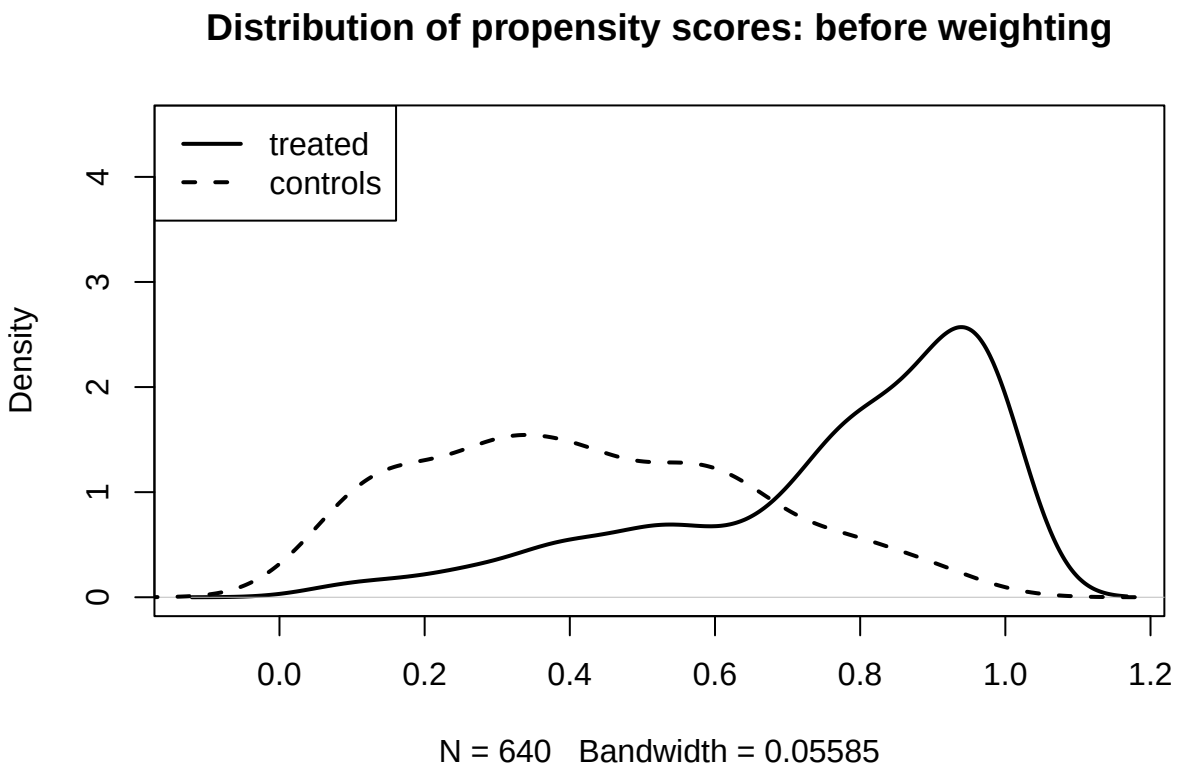
```
##
##   0   1
## 159 640
```

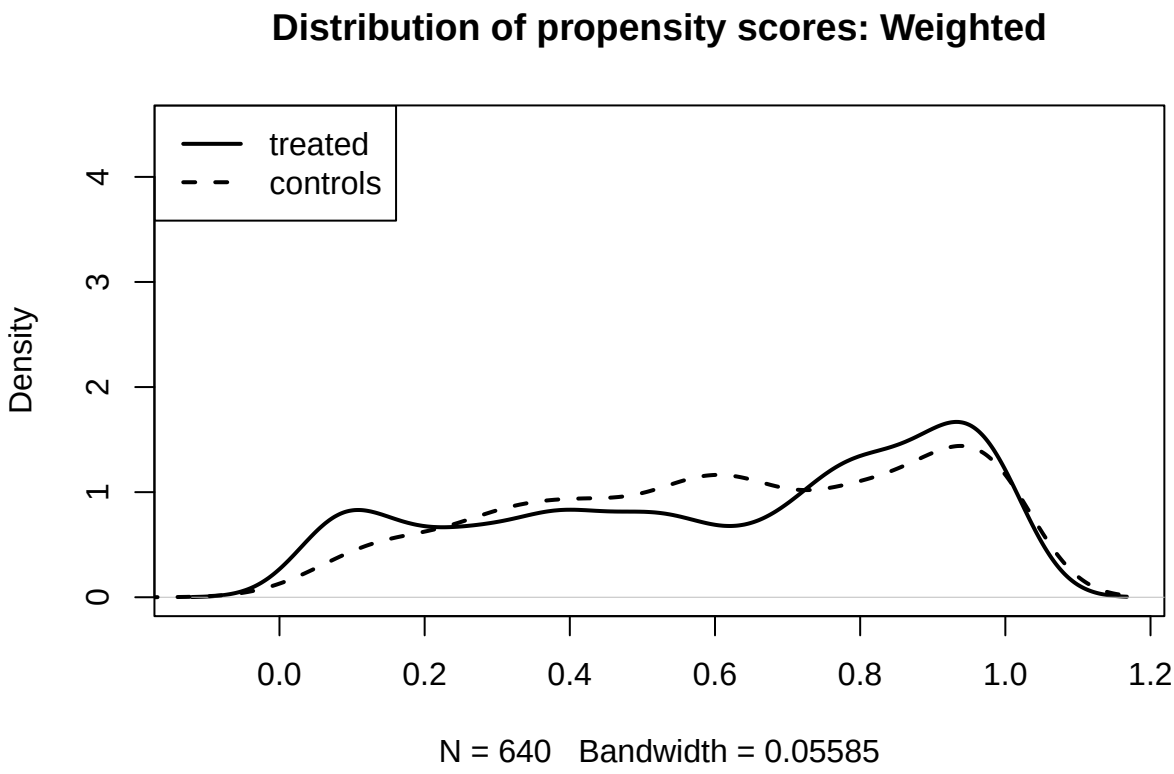


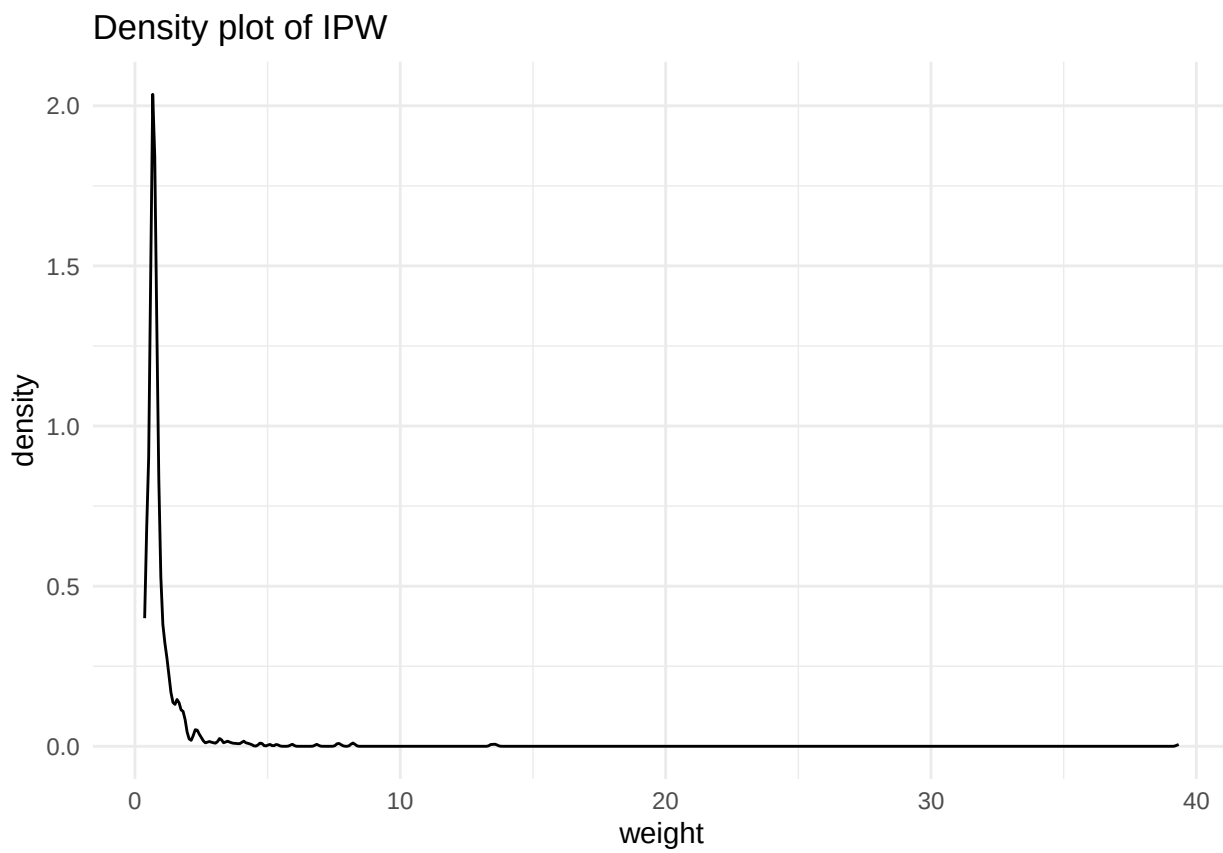
4.4 Weighting

4.4.1 Weighting propensity scores - manual version

Inverse Probability-of-Treatment Weighting (IPW):


Distribution of propensity scores: before weighting
treated
controls
Density
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
N = 640 Bandwidth = 0.05585


Distribution of propensity scores: Weighted
treated
controls
Density
0
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
1.2
N = 640 Bandwidth = 0.05585



Check how extreme the weights get / how few units are doing most of the work - it means large bias

Take regressions:

4.4.2 Weighting propensity scores - weightit() version

```
##                   Summary of weights
##
## - Weight ranges:
##
##           Min                                  Max
## treated 1.    |----|                         21.369
## control 1.005 |---------------------------| 108.116
##
## - Units with the 5 most extreme weights by group:
##
##             528    158    155    153     151
##  treated 11.996 12.146 12.889 12.939  21.369
##             315    209    186    180      11
##  control 11.668 11.984 12.912  36.94 108.116
##
## - Weight statistics:
##
##         Coef of Var   MAD Entropy # Zeros
## treated       0.935 0.442   0.218       0
## control       2.302 0.620   0.557       0
##
```

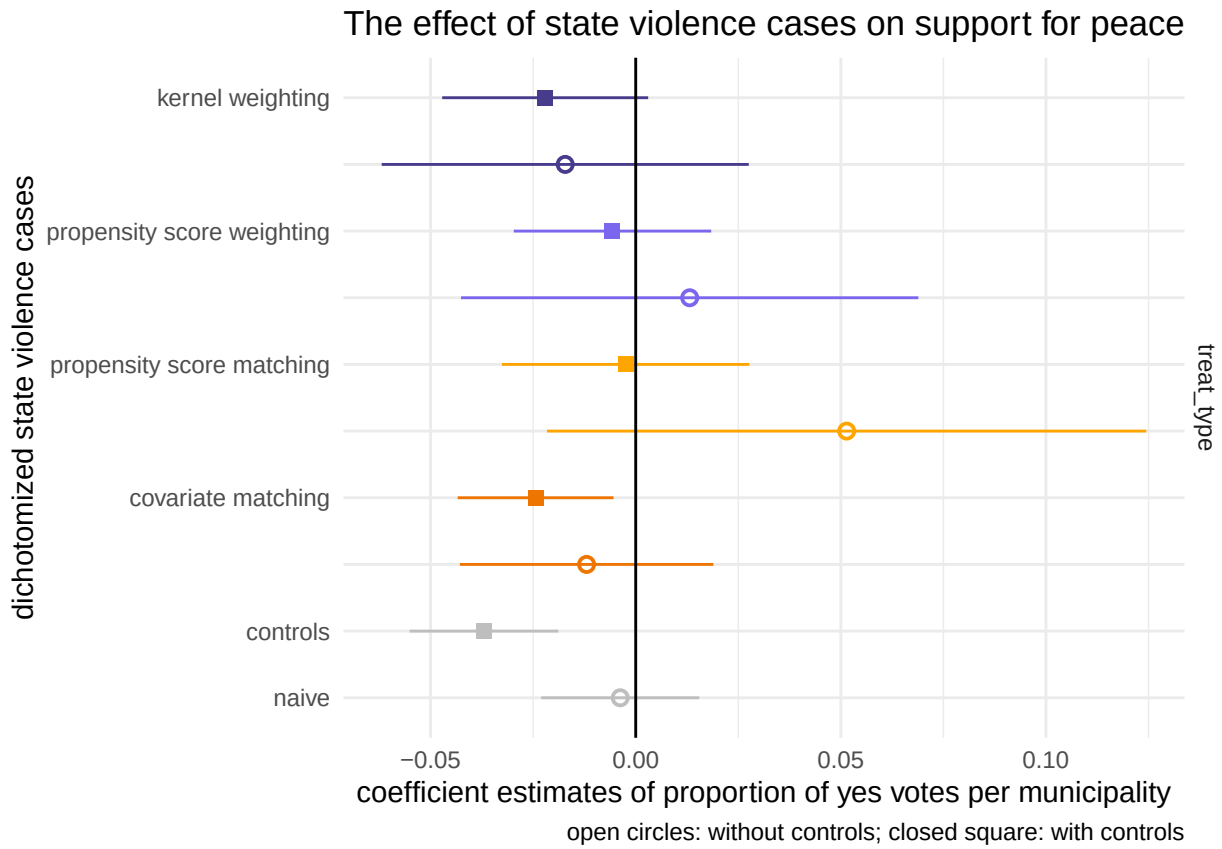
```
## - Effective Sample Sizes:
##
##            Control Treated
## Unweighted  366.    640.
## Weighted     58.24  341.72
```

5. Results

Table 4: The effect of state violence cases on support for peace

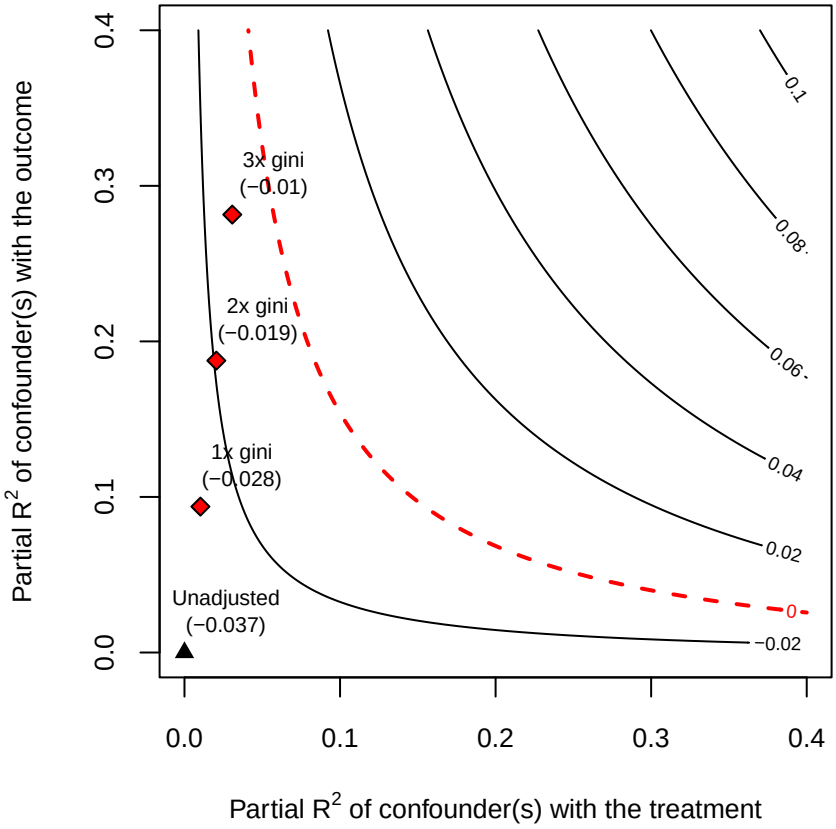
			Matching				Weighting			
	Naive	Controls	Covariates		Propensity scores				Kernel	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(Intercept)	0.496***	1.229***	0.504***	1.183***	0.441***	1.149***	0.466***	1.177***	0.509***	1.254***
	(0.008)	(0.083)	(0.014)	(0.088)	(0.037)	(0.095)	(0.028)	(0.096)	(0.022)	(0.103)
treatment	−0.004	−0.037***	−0.012	−0.024*	0.051	−0.002	0.013	−0.006	−0.017	−0.022+
	(0.010)	(0.009)	(0.016)	(0.010)	(0.037)	(0.015)	(0.028)	(0.012)	(0.023)	(0.013)
uribe		−0.400***		−0.413***		−0.438***		−0.394***		−0.450***
		(0.024)		(0.027)		(0.034)		(0.038)		(0.034)
population		−0.002		0.004		0.004		−0.005		−0.003
		(0.004)		(0.004)		(0.005)		(0.006)		(0.005)
gdp per-capita		−0.006**		−0.011***		−0.010***		−0.006**		−0.004+
		(0.002)		(0.002)		(0.002)		(0.002)		(0.002)
gini		−1.050***		−1.031***		−0.992***		−0.914***		−1.102***
		(0.132)		(0.145)		(0.156)		(0.146)		(0.166)
state		0.000		0.000		−0.002+		−0.004*		−0.002
		(0.001)		(0.001)		(0.001)		(0.002)		(0.001)
guerilla		0.000		0.000		0.000		0.000		0.000
		(0.000)		(0.000)		(0.000)		(0.000)		(0.000)
historical conflict		−0.015		−0.013		−0.013		−0.006		−0.007
		(0.015)		(0.016)		(0.019)		(0.016)		(0.017)
justice		0.020		0.016		0.020		−0.024		0.039
		(0.029)		(0.029)		(0.026)		(0.039)		(0.041)
military rank		0.053***		0.039**		0.039**		0.049**		0.043*
		(0.012)		(0.013)		(0.014)		(0.018)		(0.019)
Num.Obs.	1006	1006	854	854	799	799	1006	1006	1006	1006
R2	0.000	0.414	0.001	0.465	0.017	0.508	0.002	0.463	0.003	0.485
R2 Adj.	−0.001	0.408	0.000	0.459	0.015	0.501	0.001	0.458	0.002	0.480
RMSE	0.15	0.12					0.15	0.12	0.15	0.12
controls included	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001



6. Sensitivity analysis

Which covariate as a benchmark?



Sensitivity with regular lm() only; ipw/matching weights mess it up – how to add other versions?